

COMPANION
to

ChatGPT Teaches TikZ

Chapter 8: Coordinate Calculations (I)

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8.1 What This Companion Adds

Chapter 8 began the Intermediate part of the book by replacing measured coordinates with computed geometric relationships. This Companion develops three durable habits: calculate points from named objects, save important results with meaningful names, and test a construction by moving its original vertices.

Design

A robust geometric picture depends on a small set of original points. Every point determined by the geometry should be calculated from those points.

8.2 A Calculation Vocabulary

Form	Purpose
calc	Loads the coordinate-calculation library.
$(A)!.5!(B)$	Finds the midpoint of A and B.
$(A)!t!(B)$	Finds a fractional point from A toward B.
$(A)+(x,y)$	Translates A by a displacement.
$(A)-(x,y)$	Applies the opposite displacement.
<code>\coordinate</code>	Saves a calculated position under a name.

Load the library in the preamble:

```
\usetikzlibrary{calc}
```

8.3 The Interpolation Formula

The expression

```
 $(A)!t!(B)$ 
```

means: start at A and travel the fraction t of the way toward B. The endpoints and midpoint are special cases.

Value	Position
$t=0$	At A.
$t=0.25$	One-quarter of the way from A to B.
$t=0.5$	Halfway from A to B.
$t=0.75$	Three-quarters of the way from A to B.
$t=1$	At B.

The exclamation marks belong to the TikZ syntax; they do not mean factorials.

8.4 Computing and Naming a Midpoint

The calculation may be used directly:

```
\fill ($(A)!0.5!(B)$) circle (2pt);
```

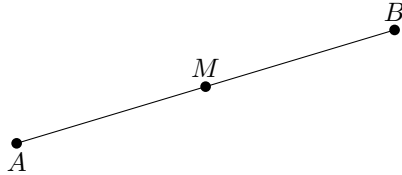
If the point will be used again, save it:

```
\coordinate (M) at ($(A)!0.5!(B)$);
```

```
\begin{tikzpicture}
  \coordinate (A) at (0,0);
  \coordinate (B) at (5,1.5);
  \coordinate (M) at ($(A)!0.5!(B)$);

  \draw (A) -- (B);
  \fill (A) circle (2pt) node[below] {$A$};
  \fill (B) circle (2pt) node[above] {$B$};
  \fill (M) circle (2pt) node[above] {$M$};
\end{tikzpicture}
```

See



Move B; the midpoint and its label move automatically.

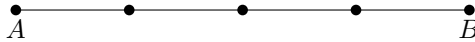
8.5 Dividing and Extending a Segment

A loop can mark several fractional points without computing any coordinates by hand.

```
\begin{tikzpicture}
  \coordinate (A) at (0,0);
  \coordinate (B) at (6,0);
  \draw (A) -- (B);

  \foreach \t in {0.25,0.5,0.75}
    \fill ($(A)!\t!(B)$) circle (2pt);
\end{tikzpicture}
```

See



The parameter need not lie between 0 and 1. A value greater than 1 continues beyond B; a negative value continues behind A.

```
\coordinate (pastB) at ($(A)!1.25!(B)$);
\coordinate (behindA) at ($(A)!-0.25!(B)$);
```

This is useful for extending lines while preserving their direction.

8.6 Translating a Point

The calculation

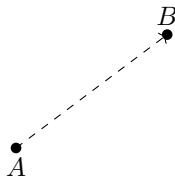
```
($A)+(1,2)$
```

starts at A, moves one unit right, and moves two units up.

```
\begin{tikzpicture}
  \coordinate (A) at (1,0.5);
  \coordinate (B) at ($A)+(2,1.5)$;

  \fill (A) circle (2pt) node[below] {$A$};
  \fill (B) circle (2pt) node[above] {$B$};
  \draw[->,dashed] (A) -- (B);
\end{tikzpicture}
```

See



Use subtraction when the displacement should be reversed.

8.7 Naming the Geometry

These two versions identify the same location:

```
\draw (A) -- ($A)!0.5!(B)$);
\draw ($A)!0.5!(B)$) -- (C);
```

```
\coordinate (midAB) at ($A)!0.5!(B)$);
\draw (A) -- (midAB);
\draw (midAB) -- (C);
```

The second version records what the point means. Names such as `midAB`, `center`, and `translatedA` turn a calculation into part of the mathematical explanation.

8.8 Worked Project: The Medial Triangle

The three midpoints of a triangle form its medial triangle.

8.8.1 Stage 1: Define the Original Vertices

```
\coordinate (A) at (0,0);  
\coordinate (B) at (5,0);  
\coordinate (C) at (1.5,3);
```

8.8.2 Stage 2: Compute the Midpoints

```
\coordinate (AB) at ($(A)!0.5!(B)$);  
\coordinate (BC) at ($(B)!0.5!(C)$);  
\coordinate (CA) at ($(C)!0.5!(A)$);
```

8.8.3 Stage 3: Draw Both Triangles

```
\draw (A) -- (B) -- (C) -- cycle;  
\draw[very thick] (AB) -- (BC) -- (CA) -- cycle;
```

8.8.4 The Complete Picture

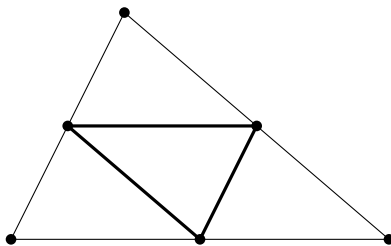
```
\begin{tikzpicture}
  \coordinate (A) at (0,0);
  \coordinate (B) at (5,0);
  \coordinate (C) at (1.5,3);

  \coordinate (AB) at ($(A)!0.5!(B)$);
  \coordinate (BC) at ($(B)!0.5!(C)$);
  \coordinate (CA) at ($(C)!0.5!(A)$);

  \draw (A) -- (B) -- (C) -- cycle;
  \draw[very thick] (AB) -- (BC) -- (CA) -- cycle;

  \foreach \P in {A,B,C,AB,BC,CA}
    \fill (\P) circle (2pt);
\end{tikzpicture}
```

See



Design

Move any original vertex. TikZ recomputes all three mid-points, so the medial triangle remains correct without any further edits.

8.9 Debugging Workshop

8.9.1 Forgetting the Dollar Signs

Incorrect:

$(A)!0.5!(B)$

Correct:

$$(A)!0.5!(B)$$

8.9.2 Forgetting the calc Library

Add `\usetikzlibrary{calc}` to the preamble.

8.9.3 Adding an Extra Pair of Parentheses

The complete coordinate is $$(A)!0.5!(B)$$. The names A and B already have their own parentheses inside the calculation.

8.9.4 Repeating a Long Calculation

If a calculated point appears twice, give it a meaningful name. This reduces typing and makes the geometry easier to audit.

8.10 Exercises

1. Draw a segment and mark its midpoint.

2. Mark the points one-third and two-thirds of the way along a segment.
3. Mark a point one-quarter of the segment length beyond B.
4. Translate a named point by $(2, 1)$.
5. Save a midpoint as M and use it in two drawing commands.
6. Construct a parallelogram by translating two vertices by the same vector.
7. Draw a triangle and compute the midpoint of each side.
8. Add the three medians to the worked project.

8.11 Solutions

8.11.1 Exercises 1–4

```
\coordinate (M) at ($(A)!0.5!(B)$);
\fill (M) circle (2pt);
```

```
\fill ($(A)!0.333!(B)$) circle (2pt);
\fill ($(A)!0.667!(B)$) circle (2pt);
```

```
\coordinate (pastB) at ($(A)!1.25!(B)$);
```

```
\coordinate (B) at ($(A)+(2,1)$);
```

8.11.2 Exercises 5–7

```
\coordinate (M) at ($(A)!0.5!(B)$);
\draw (A) -- (M);
\draw (M) -- (B);
```

```
\coordinate (C) at ($(B)+(1,2)$);  
\coordinate (D) at ($(A)+(1,2)$);  
\draw (A) -- (B) -- (C) -- (D) -- cycle;
```

Exercise 7 is the first half of the worked project: define **AB**, **BC**, and **CA** with three midpoint calculations.

For Exercise 8, connect each vertex to the midpoint of the opposite side.

8.12 ChatGPT Workshop

Ask ChatGPT to preserve the original defining points while checking every dependent calculation.

1. Rewrite this TikZ figure with calculated midpoints instead of measured midpoint coordinates.
2. Explain geometrically what $\$(A)!1.25!(B)\$$ means.
3. Find every repeated coordinate calculation and give each important point a meaningful name.
4. Check the dollar signs and parentheses in these `calc` expressions.
5. Test this construction conceptually by moving one original vertex and identify any point that would fail to update.

8.13 Final Checklist

Before continuing to Chapter 9, check that you can

1. load the `calc` library;
2. compute a midpoint and a general fractional point;
3. interpret parameter values outside the interval from 0 to 1;
4. translate a point by adding or subtracting a displacement;
5. save a useful calculation as a named coordinate;
6. distinguish original points from dependent points; and
7. verify a construction by moving an original vertex.